$$f : \mathbf{A} \to_{\langle\!|\mathbf{Set}|\!\rangle} \mathbf{B} \qquad g : \mathbf{C} \to_{\langle\!|\mathbf{Set}|\!\rangle} \mathbf{D}$$

$$(f \otimes g) : \mathbf{A} \mathbin{\overset{\circ}{\wr\!\langle}} \mathbf{C} \to_{\langle\!|\mathbf{Set}|\!\rangle} \mathbf{B} \mathbin{\overset{\circ}{\wr\!\langle}} \mathbf{D}$$

$$x \mathbin{\overset{\circ}{\wr\!\langle}} z \mapsto f(x) \mathbin{\overset{\circ}{\wr\!\langle}} g(z)$$